FSOC AI-Based Virtual Camera Tracking

Complete Technical Reference, Target Color Identification & Operating Workflow Manual

1. Executive Summary & System Purpose

The **Free-Space Optical Communications (FSOC) AI-Based Virtual Camera Tracking Simulator** is an aerospace-grade closed-loop testing platform. Mobile FSOC terminals require extreme pointing precision (micro-radians) to establish and maintain directional optical laser links between ground stations, unmanned aerial vehicles (UAVs), satellites, and aircraft. This simulator emulates a 2-axis optical gimbal system operating under harsh environmental perturbations.

The platform bridges **dynamic 3D WebGL physics**, **synthetic pinhole camera projection**, **high-speed WebSocket streaming**, **hybrid neural detection (YOLOv8 + OpenCV fallback)**, **2D Kalman filter state estimation**, and **anti-windup PID actuator control** into a unified real-time suite.

Core Operational Objectives:
- ***Rapid Coarse Alignment:*** *Detect and center dynamic optical targets within camera FOV under 3 seconds.*
- ***Occlusion & Noise Resilience:*** *Maintain trajectory tracking during temporary signal loss using Kalman prediction.*
- ***Disturbance Rejection:*** *Suppress platform vibrations, camera jitter, atmospheric turbulence, and optical blur.*
- ***Ablation & Benchmarking:*** *Evaluate system performance across 10 standardized aerospace testing scenarios.*

2. End-to-End System Workflow Architecture

The end-to-end data pipeline forms a closed loop operating at up to 60 frames per second (FPS). Below is the step-by-step breakdown of how data flows through the system components:

Step 1: 3D Scene Dynamics: The WebGL engine (Three.js / React Three Fiber) simulates 4 distinct optical beacon targets moving along user-defined trajectories (Linear, Circular, Sinewave, Random, Spiral) in 3D world space (X, Y, Z).

Step 2: Virtual Camera Projection: The Virtual Camera Model (pinhole optics) projects 3D beacon position vectors into 2D camera sensor pixel coordinates (u, v) based on current Horizontal FOV (10.0°), Vertical FOV (7.5°), and 640x480 resolution.

Step 3: Bi-Directional WebSocket Stream: The frontend transmits real-time frame telemetry to the Uvicorn FastAPI backend at `ws://localhost:8000/ws/tracking`. The backend returns state metrics, detection coordinates, and PID correction commands.

Step 4: Disturbance Engine Injection: Before AI detection, 6-DoF environmental perturbations are synthetically injected into the frame: platform vibration, CCD sensor noise, optical blur, camera jitter, atmospheric turbulence, and illumination fluctuation.

Step 5: Hybrid AI Target Detection: The backend passes frame data to YOLOv8 deep learning model. If confidence drops below threshold or GPU is unavailable, the system instantly triggers an automatic fail-safe fallback to fast OpenCV HSV color contour thresholding.

Step 6: 2D Kalman Filter Estimation: Raw pixel coordinates are processed through a 4D Kalman filter state vector [x, y, vx, vy]^T. The filter smoothes out sensor noise and predicts target position during temporary occlusions/line-of-sight loss.

Step 7: PAT State Machine Transition: The Pointing, Acquisition & Tracking (PAT) state machine evaluates error thresholds and updates the current state (SEARCHING → DETECTED → ACQUIRING → TRACKING →

LOCKED → TARGET_LOST).

Step 8: Discrete Anti-Windup PID Control: Dual-axis discrete PID controllers compute proportional, integral, and derivative angular velocity commands for Yaw (Pan) and Pitch (Tilt) gimbals with anti-windup clamping to prevent actuator saturation.

3. Target Beacon & HUD Color Coding Guide

To ensure clear visual distinction during complex multi-target operations and telemetry debugging, specific colors have been assigned to 3D beacon targets, camera overlays, HUD crosshairs, and system status indicators:

Target / Element Name	Color Name & Hex Code	Role & Description
BEACON-01 (Primary Target)	**Emerald Green** #10B981	**Default Primary Beacon** under active closed-loop tracking. Represents the main optical laser transmitter.
BEACON-02 (Secondary Target)	**Azure Blue** #3B82F6	Auxiliary target for multi-beacon acquisition and target switching stress tests.
BEACON-03 (Tertiary Target)	**Amber Yellow** #F59E0B	Interference target / solar glare decoy used to test background false positive rejection.
BEACON-04 (Quaternary Target)	**Hot Pink** #EC4899	High-velocity dynamic target for testing rapid gimbal slewing limits.
Optical Boresight Reticle	**Muted Red** #EF4444 (20% Opacity)	Center crosshair & alignment circle on camera feed representing zero-error boresight center (320, 240).
AI Detector Bounding Box	**Royal Blue Box** #3B82F6	Surrounds detected beacon in camera feed with live confidence tag (e.g. BEACON 0.95) and raw center dot.
Kalman Predicted Target	**Pulsating Emerald** #10B981 / #34D399	Pulsating green ring and dot indicating filtered/predicted target position used by the PID controller.
Angular Error Vector Line	**Dotted Green Line** #34D399	Dashed vector connecting central boresight reticle to target pixel, representing current tracking error.
Fallback Warning Badge	**Pulsating Amber** #F59E0B	Animated warning pill displayed on HUD when system switches from YOLOv8 to OpenCV HSV thresholding.

4. Pointing, Acquisition & Tracking (PAT) State Transitions

The system enforces a robust finite state machine to manage state transitions and avoid rapid oscillation (chattering) between states:

1. **SEARCHING:** Default initial state. The gimbal executes a pre-programmed raster scan pattern (Speed: 10°/s, Span: ±20° H / ±15° V) until a beacon enters the optical Field of View.
2. **DETECTED:** Triggered immediately when AI detector detects target pixels within camera FOV. Initializes tracking candidate counter.
3. **ACQUIRING:** Transitioned when target is consistently detected across 3 consecutive frames. Kalman filter initializes state vector [x, y, 0, 0]^T.

4. **TRACKING:** Active closed-loop control. Discrete PID controller actively drives Pan and Tilt gimbals to align target with optical boresight center (320, 240).
5. **LOCKED:** Achieved when Total Angular Error remains below 0.15° continuously for at least 0.5 seconds. Represents optical communication link lock.
6. **TARGET_LOST:** Triggered when beacon is occluded or lost. Kalman filter predicts trajectory for up to 3.0 seconds. If re-detected within 3s, system returns to TRACKING; otherwise returns to SEARCHING.

5. Technical Component Architecture & Tech Stack

The software architecture is split into high-performance Frontend and Backend subsystems designed for zero-lag real-time execution:

Subsystem	Technologies & Libraries	Key Modules & Responsibilities
Frontend UI & 3D Simulation	React 18, Vite, Three.js, React Three Fiber, TailwindCSS, Zustand, Lucide Icons, Recharts	• Dashboard.jsx: Core application container. • SimulationScene.jsx: 3D WebGL Canvas. • GimbalCamera.jsx: 2-Axis gimbal mount model. • Beacon.jsx: Multi-beacon spatial renderer. • CameraFeed.jsx: Real-time HUD overlay. • TelemetryPanel.jsx: Real-time charts. • ControlPanel.jsx: Interactive tuning sliders. • useSimulationStore.js: Global state store.
Backend Engine & AI Analytics	Python 3.10+, FastAPI, Uvicorn, WebSockets, OpenCV (cv2), Ultralytics YOLOv8, NumPy, SciPy, ReportLab, Matplotlib	• main.py: FastAPI REST & WebSocket server. • camera/model.py: Pinhole camera math. • detection/detector.py: YOLO / HSV detector. • detection/disturbances.py: 6-DoF disturbance engine. • tracking/kalman.py: 2D linear Kalman filter. • tracking/state_machine.py: PAT FSM transition logic. • tracking/metrics.py: Benchmark metrics logging. • experiments/manager.py: Scenario evaluator. • experiments/reports.py: PDF report compiler.

6. Control & Estimation Math Specifications

A. Discrete Anti-Windup PID Controller Equations:

The gimbal pan and tilt angular error signals e_pan(t) and e_tilt(t) are processed via discrete PID control with anti-windup clamping to prevent integrator saturation during large step changes:

```
u(t) = K_p · e(t) + K_i · Σ [e(t) · Δt] + K_d · [(e(t) - e(t - Δt)) / Δt]
```

Default Gain Constants: Proportional K_p = 2.5, Integral K_i = 0.05, Derivative K_d = 0.15. Max actuator speed limit: ±180.0 °/s.

B. 2D Kalman Filter State Representation:

State vector X = [x, y, v_x, v_y]^T. The state transition matrix F assumes constant velocity motion model between time steps Δt:

```
F = [[1, 0, Δt, 0], [0, 1, 0, Δt], [0, 0, 1, 0], [0, 0, 0, 1]]
```

7. Operating Manual & User Execution Workflow

Follow these step-by-step instructions to run, operate, test, and evaluate the FSOC Virtual Camera Tracking Simulator:

Step 1: Start Backend Server:

Open Terminal 1 in the project directory:
cd backend
.venv\Scripts\activate (Windows) or source .venv/bin/activate (Linux/Mac)
python -m uvicorn app.main:app --reload
Server starts on http://127.0.0.1:8000 with WebSocket active.

Step 2: Start Frontend Application:

Open Terminal 2 in the project directory:
cd frontend
npm run dev
Open http://localhost:5173 in Google Chrome or Microsoft Edge.

Step 3: Operating Dashboard & Target Selection:

- **Play / Pause:** Toggle simulation clock playback.
- **Target Beacon Selection:** Select active beacon (BEACON-01 Green, BEACON-02 Blue, BEACON-03 Amber, BEACON-04 Pink).
- **Trajectory Mode:** Switch target motion profiles (Linear, Circular, Sinewave, Random, Spiral).
- **Target Speed Slider:** Adjust velocity from 0.5 m/s up to 20.0 m/s.

Step 4: Disturbances & Environmental Stress Testing:

In the **Control Panel**, adjust disturbance sliders to evaluate system robustness:
- **Platform Vibration:** High-frequency mechanical jitter (0 - 100%).
- **Camera Jitter:** Structural wind buffeting.
- **Image Noise / Blur:** Atmospheric haze & CCD thermal noise.
- **Atmospheric Turbulence:** Optical refraction fluctuations.

Step 5: Ablation Studies & Feature Toggles:

Toggle individual system components on/off to isolate performance impact:
- **AI Detection Toggle:** Disables YOLO/HSV detector (forces open-loop mode).
- **Kalman Filter Toggle:** Disables state smoothing & occlusion prediction.
- **PID Controller Toggle:** Disables active gimbal correction.
- **Disturbances Toggle:** Disables environmental perturbations.

Step 6: Running Benchmarks & Generating Reports:

- Navigate to **Experiment Benchmarks Tab**.
- Select any of 10 aerospace scenarios (e.g. High Jitter, Dense Occlusion, Supersonic Slew).
- Click **Run Scenario** to initiate automated evaluation.
- Click **Export PDF Report** to download a formal 2-page evaluation document with Matplotlib charts.

8. Predefined Aerospace Benchmark Scenarios

The platform includes 10 standardized test scenarios to validate tracking precision under stress:

ID	Scenario Name	Test Purpose & Configuration
EXP-01	Baseline Nominal Tracking	Zero disturbances, linear trajectory at 5.0 m/s. Establishes baseline PID response.
EXP-02	High Mechanical Jitter	Vibration = 80%, Jitter = 70%. Tests Kalman filter noise suppression capabilities.
EXP-03	Severe Atmospheric Blur	Blur = 90%, Image Noise = 60%. Tests OpenCV color thresholding & YOLO robustness.
EXP-04	Dense Signal Occlusion	Beacon occluded for 2.5s intervals. Tests Kalman trajectory velocity prediction.
EXP-05	Supersonic Target Slew	Target speed = 25.0 m/s with sharp 90° direction changes. Tests PID slewing response.
EXP-06	Multi-Beacon Interference	BEACON-03 (Amber) crosses path of BEACON-01 (Green). Tests decoy rejection.
EXP-07	Low Illumination Solar Dusk	Illumination variation = 95%. Tests optical sensor gain & exposure handling.
EXP-08	Raster Search Recovery	Target starts 40° out of FOV. Tests automatic raster search scan pattern efficiency.
EXP-09	Full Environmental Stress	All 6 disturbance sliders at 75%. Evaluates overall system break-lock limits.
EXP-10	Ablation Baseline (No Kalman)	Kalman filter disabled under jitter. Quantifies exact RMSE degradation without Kalman.

9. Verification & Conclusion

This project successfully demonstrates a complete end-to-end Free-Space Optical Communications tracking simulator. By combining deep learning AI, state estimation, and physical gimbal dynamics, the system achieves sub-pixel tracking accuracy and resilient closed-loop lock retention under extreme environmental disturbances.